

सेफ्टी अलर्ट / SAFETY ALERT

OISD/SA/2024-25/MOPOL/04

Dt.10/05/2024

INTRODUCTION

Title: Pipeline Pilferage and Fire incident in 16" MS Jetty Pipeline

Location: Pipeline corridor outside terminal and Refinery premises

Date & Time: 18.02.2024 at approx. 02:38 hrs.

Loss/ Outcome: 4 KL product loss.

Injury/ Fatality: Nil

INCIDENT

- ❖ A fire incident occurred at approx. 02:38 hrs on 16" MS industry Jetty Pipeline outside refinery boundary wall.
- ❖ A network of pipelines (MS, HSD, SKO & FO) is used on industry basis for coastal receipt and PLT operations.
- ❖ No operation being carried out in the MS line at the time of the incident and the valves were closed at both ends i.e. Oil Jetty and OMC terminals.
- ❖ The control room shift officer at the Refinery Terminal spotted fire outside its boundary and raised fire alarm. The ongoing operations in adjacent OMC terminals were suspended.
- ❖ Initially, the Refinery tried to extinguish the fire by using WCFMs from inside the boundary wall. However, the fire could not be extinguished due to some difficulties faced by the combat team.
- ❖ The refinery thus, deployed the fire tender at the incident site for combating the fire. However, the fire tender could not be suitably positioned to cover the incident site with water throw due to a difficult terrain and muddy land.
- ❖ Subsequently, the incident was declared as Level-II contingency and help was sought from Mutual aid members by Refinery. The fire Tenders of Dock Complex and State Fire Service were deployed at the spot and other industry members were put on readiness.
- ❖ The fire fighting continued for approx. three hours and the fire was extinguished at 05:26 hrs. Foam-water application was continued for another 15-20 minutes to prevent any flashback or delayed ignition.
- ❖ It was observed that ½" hole at the 6-o'clock position of the MS line was drilled. 10 nos. of 50-liter cap. jerry cans filled with MS were found at the site. The pipeline was repaired by fixing a clamp by the Refinery.
- ❖ OMC custodian of pipeline registered a complaint with the police station.
- ❖ The various items such as handheld cordless power drilling machines, spanners, pliers, fabricated fixtures, fittings for tapping in the pipeline, molten jerry can etc. were recovered from the incident site and handed over to police.

OBSERVATIONS

- ❖ As per OMC investigation report, pilferage was being carried out by miscreants evidenced by presence of handheld battery-operated drilling machine and fittings for tapping of the pipeline at the site.
- ❖ There was a pool of products formed which caught fire during ongoing pilferage by miscreants and the vegetation around the pipelines also caught fire.

- ❖ The incident spot is around 40m away from motorable road separated by thick vegetation/muddy soil due to which the location is inaccessible for firefighting from roadside.
- ❖ Security guards were not walking along the pipelines during patrolling, they were walking on the road which is approx. 40 mts from the pipelines. Due to dense vegetation along pipelines, there is no clear visibility from this distance to detect suspicious activities especially during night hours.
- ❖ There are four OMC terminals and one refinery located adjacent to each other, but no common communication system among the OMC installations available.
- ❖ There was no CCTV coverage of the network of pipelines.
- ❖ Approximately 4KL product loss due to pilferage and fire.
- ❖ There were no injuries or fatalities.



ROOT CAUSE

Pilferage/miscreant activities: Continuous and uncontrolled flow of MS from pipeline during pilferage formed flammable atmosphere and caught fire.

Pipeline Patrolling: Ground patrolling by walking along the pipeline was not carried out by line walkers.

Poor Maintenance and Inspection: Poor upkeeping of the pipeline corridor leading to growth of intergrown tress/ vegetations obstructing clear visibility.

RECOMMENDATIONS

- ❖ Security patrolling pathway alongside of the network of pipelines for effective monitoring.
- ❖ The pipeline corridor shall be free from dense vegetation.
- ❖ Explore fixing PTZ CCTV cameras to monitor vulnerable sections of pipelines.
- ❖ OMC industry members should devise a plan to secure the pipeline corridor area.
- ❖ All the nearby installations should explore a common emergency communication system.

Provided for information purposes only. This information should be evaluated to determine if it is applicable in your operations, to avoid recurrence of such incidents.